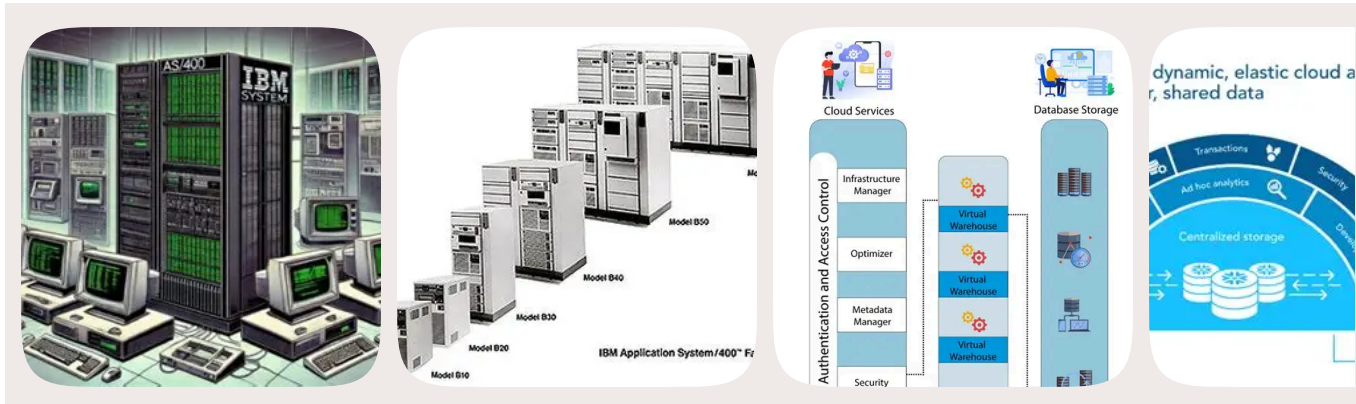


AS/400 → Snowflake Migration Playbook (Enterprise Edition)



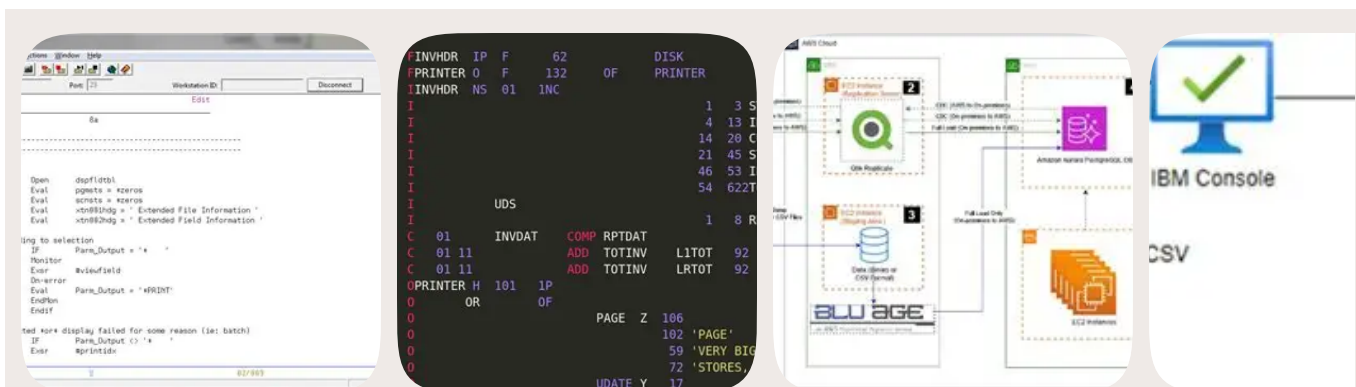
1. Program Foundations & Readiness

Objective: Establish governance, scope, and architecture before touching data.

- Migration charter — business drivers, scope, SLAs, compliance.
- Environment readiness — Snowflake accounts, roles, warehouses, stages.
- Landing zone setup — S3/ADLS/GCS buckets, encryption, IAM.
- Connectivity validation — ODBC/JDBC, firewall rules, VPN/Direct Connect/ExpressRoute.
- Data classification — PII, PCI, mortgage-sensitive fields.

Deliverables: Architecture blueprint, RACI, security model, migration waves.

2. Legacy System Assessment (AS/400 / IBM i)

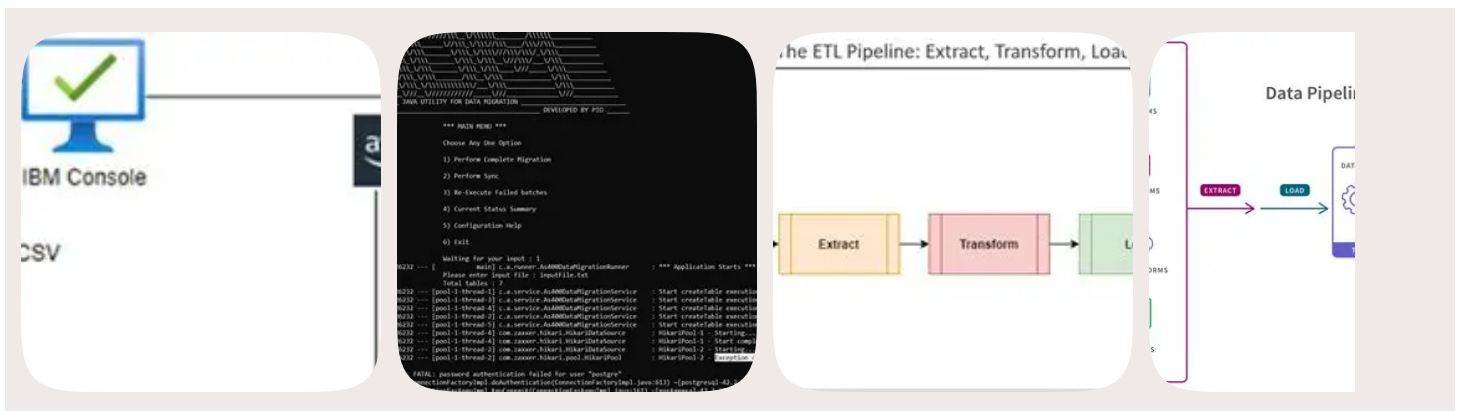


Objective: Reverse-engineer the legacy system to understand data, logic, and dependencies.

- Inventory PF/LF files — physical/logical files, DDS, constraints.
- RPG III/IV logic analysis — embedded business rules, date logic, calculations.
- Job flow mapping — CL programs, nightly cycles, dependencies.
- Data profiling — packed decimals, zoned fields, EBCDIC, nullability.
- CDC feasibility — journaling, triggers, timestamp columns.

Deliverables: Source-to-target mapping baseline, RPG logic catalog, dependency map.

3. Extraction Strategy (Full + Incremental)



Objective: Extract data from AS/400 with fidelity, auditability, and repeatability.

Extraction Methods

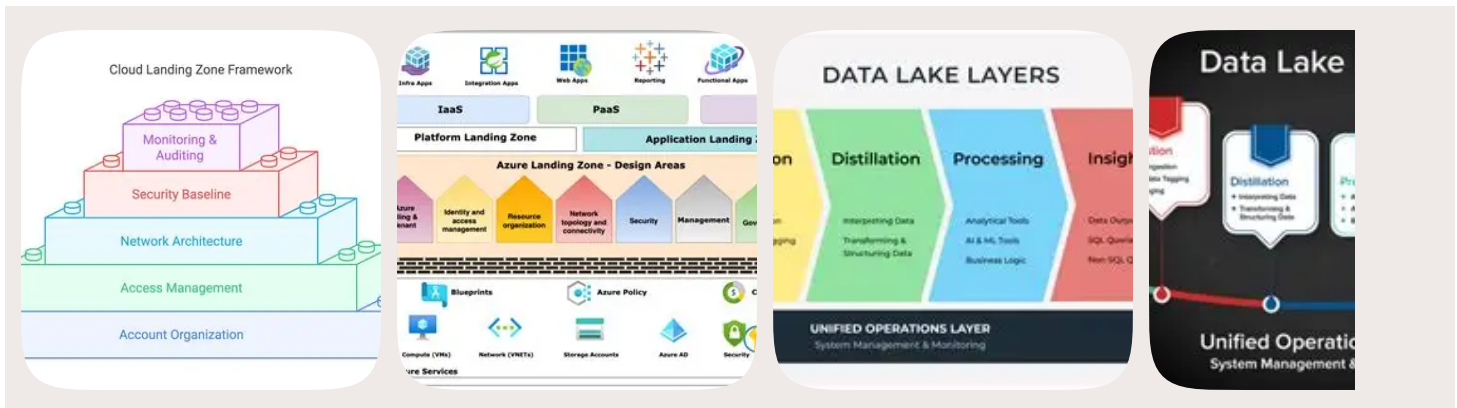
- Direct DB2/400 extraction — ODBC/JDBC, CPYTOIMPF, RUNSQLSTM.
- Flat-file exports — fixed-width, EBCDIC → UTF-8 conversion.
- Journal-based CDC — incremental loads for cutover.
- API/service extraction — when business logic must be preserved.

Extraction Best Practices

- Preserve 1:1 source fidelity (no transformations).
- Include metadata files (row counts, checksums, timestamps).
- Use parallel unloads for large PFs.
- Automate retry + checkpointing.

Deliverables: Extract scripts, CDC design, raw data samples, metadata framework.

4. Cloud Landing Zone & Staging Architecture



Objective: Land AS/400 data in cloud storage with full lineage.

Landing Zone Layers

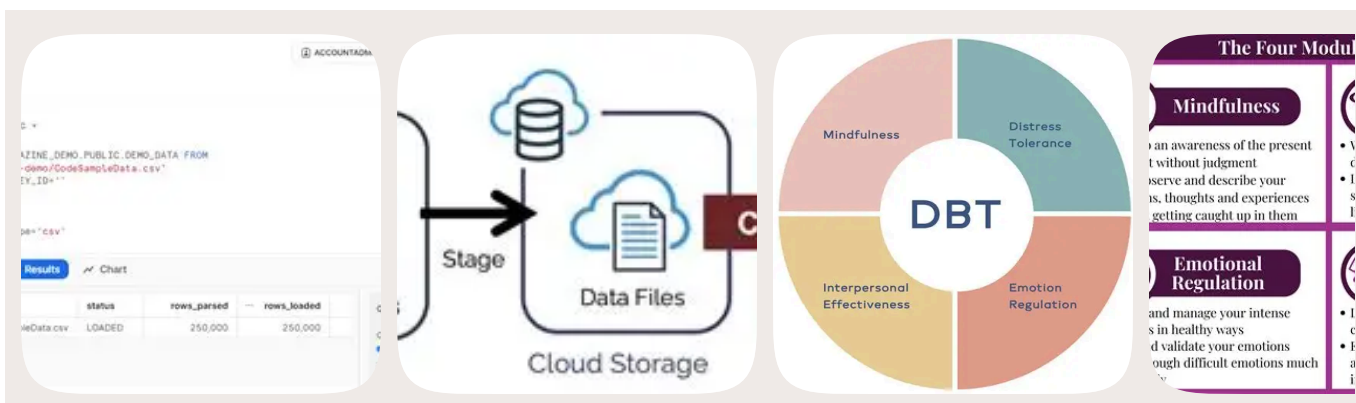
- **Raw Zone** — exact AS/400 extracts, no changes.
- **Normalized Zone** — packed → numeric, EBCDIC → UTF-8.
- **Validated Zone** — schema-checked, row-count validated.

Snowflake Integration

- External stages (S3/ADLS/GCS).
- File formats (CSV, PARQUET, FIXED-WIDTH).
- COPY INTO with validation options.

Deliverables: Landing zone folder structure, IAM roles, Snowflake stages.

5. Snowflake Ingestion & ELT Framework



Objective: Load raw AS/400 data into Snowflake and prepare it for transformation.

Ingestion Patterns

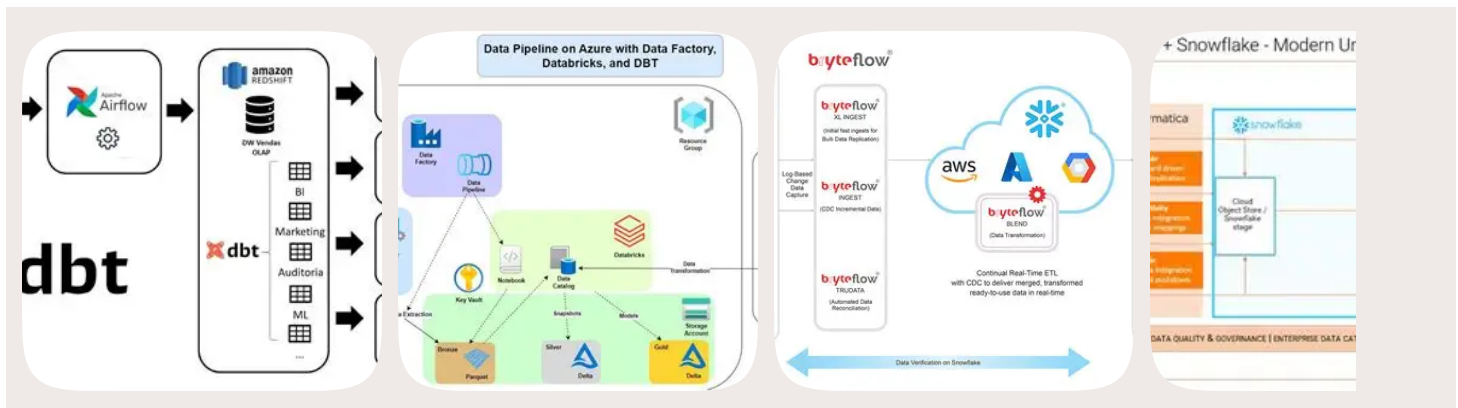
- Snowpipe — continuous ingestion.
- COPY INTO — batch ingestion.
- Streams & Tasks — incremental processing.
- Dynamic Tables — automated ELT refresh.

Key Considerations

- Use **VALIDATION_MODE** to detect malformed rows.
- Use **AUTO-INGEST** for real-time CDC.
- Partition large tables by **date**, **region**, or **account**.

Deliverables: COPY scripts, Snowpipe configs, ingestion monitoring dashboards.

6. Transformation Layer (DBT + Snowflake ELT)



Objective: Rebuild AS/400 business logic using DBT and Snowflake SQL.

DBT Model Layers

- Staging models — column renaming, type casting.
- Intermediate models — joins, business logic.
- Mart models — analytics-ready structures.

Rebuilding RPG Logic

- Convert RPG calculations → SQL expressions.
- Convert CL job flows → DBT DAG dependencies.

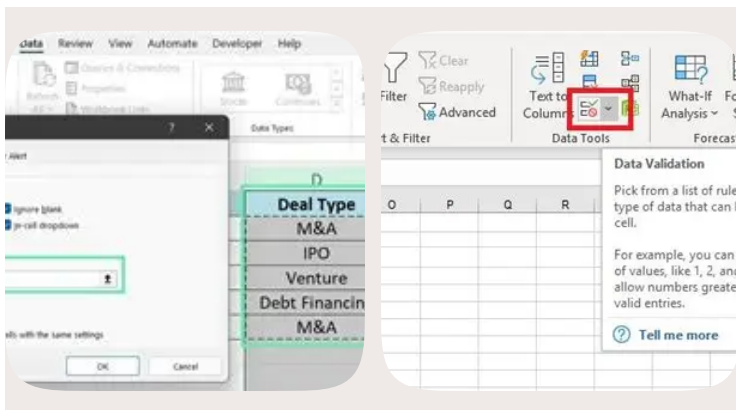
- Convert PF/LF relationships → Snowflake constraints + joins.

Performance Optimization

- Micro-partition pruning.
- Clustering keys.
- Result caching.
- Warehouse sizing.

Deliverables: DBT project, lineage graph, transformation specs.

7. Validation & Reconciliation



Objective: Ensure migrated data is accurate, complete, and audit-ready.

Validation Types

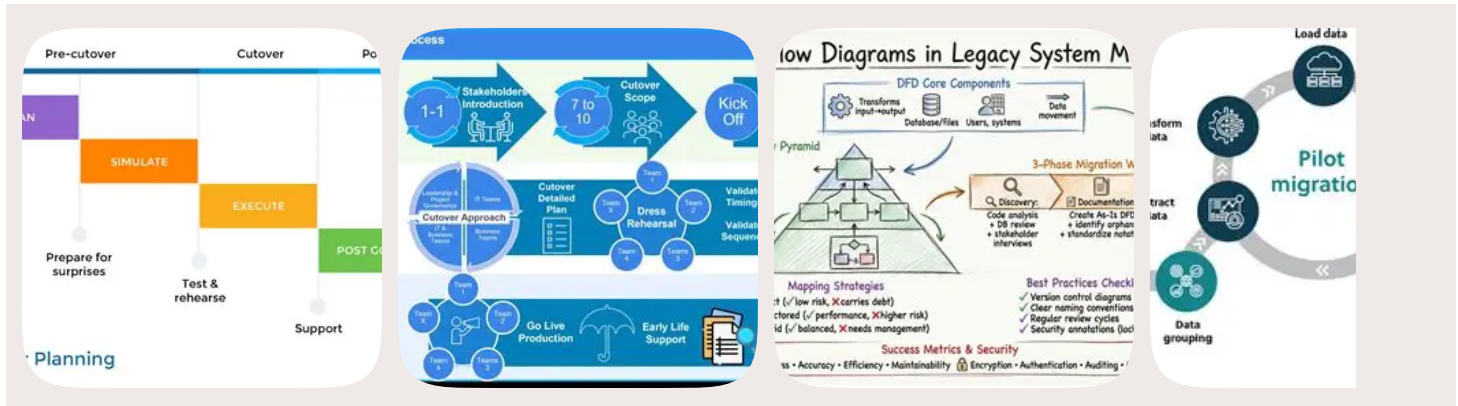
- Schema validation — column counts, types.
- Metric validation — row counts, sums, distinct counts.
- Row-level validation — required for mortgage data.
- Functional validation — compare legacy vs Snowflake outputs.

Tools

- DBT tests.
- Snowflake EXCEPT queries.
- Hash-based reconciliation.

Deliverables: Validation reports, reconciliation dashboards, audit logs.

8. Cutover Strategy



Objective: Transition from AS/400 to Snowflake with minimal disruption.

Cutover Steps

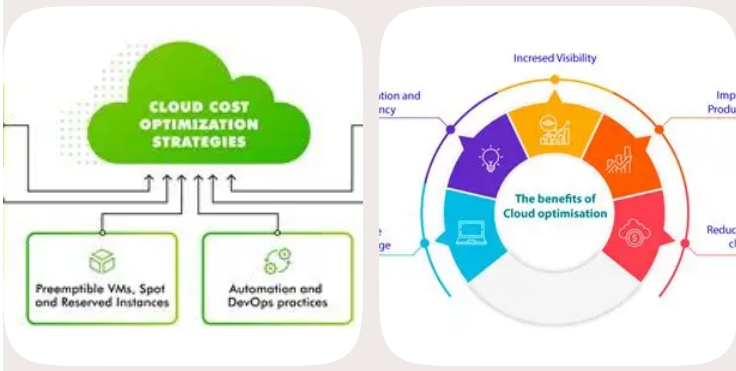
- Freeze window.
- Final CDC replay.
- Parallel run (legacy + Snowflake).
- Business sign-off.
- Switch downstream systems to Snowflake.

Cutover Models

- **Big Bang** — single cutover event.
- **Phased** — migrate subject areas in waves.
- **Hybrid** — combine both.

Deliverables: Cutover runbook, rollback plan, communication plan.

9. Decommissioning & Optimization



Objective: Retire legacy systems and optimize Snowflake for long-term use.

Decommissioning

- Archive AS/400 data.
- Retire PF/LF files, RPG jobs, CL programs.
- Shut down hardware or LPARs.

Snowflake Optimization

- Warehouse right-sizing.
- Storage tiering.
- RBAC/ABAC governance.
- Cost monitoring (Query History, Resource Monitors).

Deliverables: Decommission checklist, cost optimization plan.

10. End-to-End Architecture Diagram (Conceptual)

